

Function Operations, Domain & Range, and Inverse Functions

Calculus Honors (US)

Topics: Function Composition | Domain and Range | Inverse Functions (Calculation and Verification)

1 Domain and Range

Definitions

The **domain** of a function is the complete set of possible x -values (inputs). The **range** is the complete set of possible y -values (outputs).

Note: A square root expression $\sqrt{u}$ requires $u \geq 0$ (you cannot take the square root of a negative real number), and the result $\sqrt{u}$ itself is always ≥ 0 .

Example: Finding the Domain and Range of $f(x) = \sqrt{1 - 2x}$

Domain: the expression under the root must be non-negative:

$$1 - 2x \geq 0 \Rightarrow 1 \geq 2x \Rightarrow x \leq \frac{1}{2} \Rightarrow \text{Domain} = (-\infty, \frac{1}{2}].$$

Range: since a square root is always ≥ 0 , and $\sqrt{1 - 2x} \rightarrow \infty$ as $x \rightarrow -\infty$:

$$\text{Range} = [0, \infty).$$

2 Function Composition

Definition

The **composition** of f and g , written $(f \circ g)(x)$, means applying g first and then f :

$$(f \circ g)(x) = f(g(x)).$$

In general, $(f \circ g)(x) \neq (g \circ f)(x)$ – order matters.

Example

Let $f(x) = \sqrt{1-2x}$ and $g(x) = \frac{x}{2} - 1$.

$$(f \circ g)(x) = f(g(x)) = \sqrt{1-2\left(\frac{x}{2} - 1\right)} = \sqrt{1-x+2} = \sqrt{3-x}$$

$$(g \circ f)(x) = g(f(x)) = \frac{\sqrt{1-2x}}{2} - 1$$

3 Inverse Functions

Definition and Method

The **inverse** of f , written f^{-1} , “undoes” f : if $f(a) = b$, then $f^{-1}(b) = a$. To find $f^{-1}(x)$ algebraically:

1. Write $y = f(x)$.
2. Swap x and y .
3. Solve the new equation for y . The result is $f^{-1}(x)$.

Example (Linear Function)

Find the inverse of $f(x) = 1 + x$.

$$y = 1 + x \xrightarrow{\text{swap}} x = 1 + y \Rightarrow y = x - 1$$

$$\therefore f^{-1}(x) = x - 1.$$

4 Finding the Inverse of a More Complex Function

Example

Find the inverse of $f(x) = \sqrt{1-2x}$.

$$y = \sqrt{1-2x} \xrightarrow{\text{swap}} x = \sqrt{1-2y} \Rightarrow x^2 = 1-2y \Rightarrow y = \frac{1-x^2}{2}$$

$$\therefore f^{-1}(x) = \frac{1-x^2}{2}.$$

Restricting the Domain of the Inverse

Since f has domain $(-\infty, \frac{1}{2}]$ and range $[0, \infty)$, the inverse must have its domain and range *swapped* from the original:

$$\text{Domain of } f^{-1} = [0, \infty), \quad \text{Range of } f^{-1} = (-\infty, \frac{1}{2}].$$

Note: The source notes state the domain of f^{-1} as all real numbers, $(-\infty, \infty)$. Algebraically, the expression $\frac{1-x^2}{2}$ is defined for every real x , but as the inverse of this particular f , its domain must be restricted to match f 's range, $[0, \infty)$ – otherwise f^{-1} would not correctly undo f . The restricted domain is used above.

5 Completing the Square (Vertex Form)

Vertex Form

Any quadratic expression $ax^2 + bx + c$ can be rewritten in **vertex form**

$$f(x) = a(x - h)^2 + k,$$

where (h, k) is the vertex of the parabola. If $a < 0$, the parabola opens **downward** and k is the maximum value of f ; if $a > 0$, it opens **upward** and k is the minimum value.

Example: Rewriting $f(x) = \frac{1 - x^2}{2}$ in Vertex Form

$$f(x) = \frac{1 - x^2}{2} = -\frac{x^2 - 1}{2} = -\frac{1}{2}x^2 + \frac{1}{2} = -\frac{1}{2}(x - 0)^2 + \frac{1}{2}$$

So $a = -\frac{1}{2} < 0$ (downward-opening) with vertex $(h, k) = (0, \frac{1}{2})$, confirming the maximum value $\frac{1}{2}$ and range $(-\infty, \frac{1}{2}]$ found above.

Note: The source notes factor $1 - x^2$ as $-(x^2 + 1)$, which is an algebra slip – the correct factoring is $1 - x^2 = -(x^2 - 1)$. That error would have given vertex $(0, -\frac{1}{2})$ and range $(-\infty, -\frac{1}{2}]$, which contradicts the domain of the original f found in Section 1. The corrected version above is consistent with that earlier result.

6 Verifying an Inverse Function

Verification Method

To confirm that f^{-1} is truly the inverse of f , check that composing them (in the appropriate domain) returns the input:

$$(f \circ f^{-1})(x) = x \quad \text{and} \quad (f^{-1} \circ f)(x) = x.$$

Verifying $f(x) = \sqrt{1 - 2x}$ and $f^{-1}(x) = \frac{1 - x^2}{2}$

For $x \geq 0$ (the domain of f^{-1}):

$$(f \circ f^{-1})(x) = \sqrt{1 - 2\left(\frac{1 - x^2}{2}\right)} = \sqrt{1 - (1 - x^2)} = \sqrt{x^2} = x \quad \checkmark$$

For $x \leq \frac{1}{2}$ (the domain of f):

$$(f^{-1} \circ f)(x) = \frac{1 - (\sqrt{1 - 2x})^2}{2} = \frac{1 - (1 - 2x)}{2} = \frac{2x}{2} = x \quad \checkmark$$

Both checks return x , confirming $f^{-1}(x) = \frac{1 - x^2}{2}$ (restricted to $x \geq 0$) is the correct inverse of $f(x) = \sqrt{1 - 2x}$.